

EXECUTIVE INTERNSHIP SUMMARY REPORT

BRAHMA Workspace Core Projects & Systems Architecture

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Designation:	Software Engineering Intern
Reporting To:	Amteshwar Raj Singh
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TECHNICAL PROJECT PORTFOLIO

■ Project BHOOMI — Agriculture Intelligence Ecosystem

Objective/Focus: Empower farmers with data-driven advisory, hyper-local weather alerts, pest/disease detection through image classification, and real-time market pricing intelligence.

Current Status: Active design phase for multi-agent validation protocols. Implementing Human-in-the-loop (HITL) checkpoints to ensure the accuracy of advisory outputs before dissemination.

■ Project SMRITI — Personal AI Second Brain (RAG Vault)

Objective/Focus: Build a semantic search and RAG retrieval vault for notes, PDFs, web links, and voice notes.

Current Status: Django REST Framework backend structure setup is complete. Configured database schema with PostgreSQL + pgvector, and initialized Celery + Redis workers for asynchronous Whisper transcription and OCR jobs.

■ Project NEXUS_CELL — Intern Diagnostics & Prototyping Hub

Objective/Focus: A unified responsive web interface showcasing intern progress. Integrates Deepanshu's CRUD Expense & Task tracker and Rudraksh's Talent OS FAQ bot prototype.

Current Status: MVP dashboard design completed using a modern dark-theme UI. Moving toward REST API integrations and local storage state persistence.

■ Projects NAARTH & ARADH — Governance & Task Allocation

Objective/Focus: NAARTH defines role-based responsibility mapping (Owner, Reviewer, Approver) to avoid delivery gaps. ARADH handles dynamic task allocation and load balancing between humans and AI agents.

Current Status: Currently migrating legacy RBAC and workload systems to dynamic responsibility-mapping data models.

■ Process Automation & Email Dispatcher

Objective/Focus: Automated delivery system for status reporting, alerts, and PDF report delivery using secure SMTP connection handling.

Current Status: Fully integrated environment-variable configuration (.env) for credential isolation. Configured unified public and private log systems to ensure end-to-end auditability.

I welcome the opportunity to discuss these architectural updates, development workflows, and our next milestone plans at your convenience. Thank you for your continued guidance and mentorship.

Sincerely,

Anchal Koli

Software Engineering Intern - BRAHMA Workspace

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